Image Features:

Scale Invariant Interest Point Detection

Scale Invariant Interest Points

How can we **independently** select interest points in each image, such that the detections are repeatable across different scales?

[Source: K. Grauman, slide credit: R. Urtasun]

Scale Invariant Interest Points

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Scale Invariant Interest Points

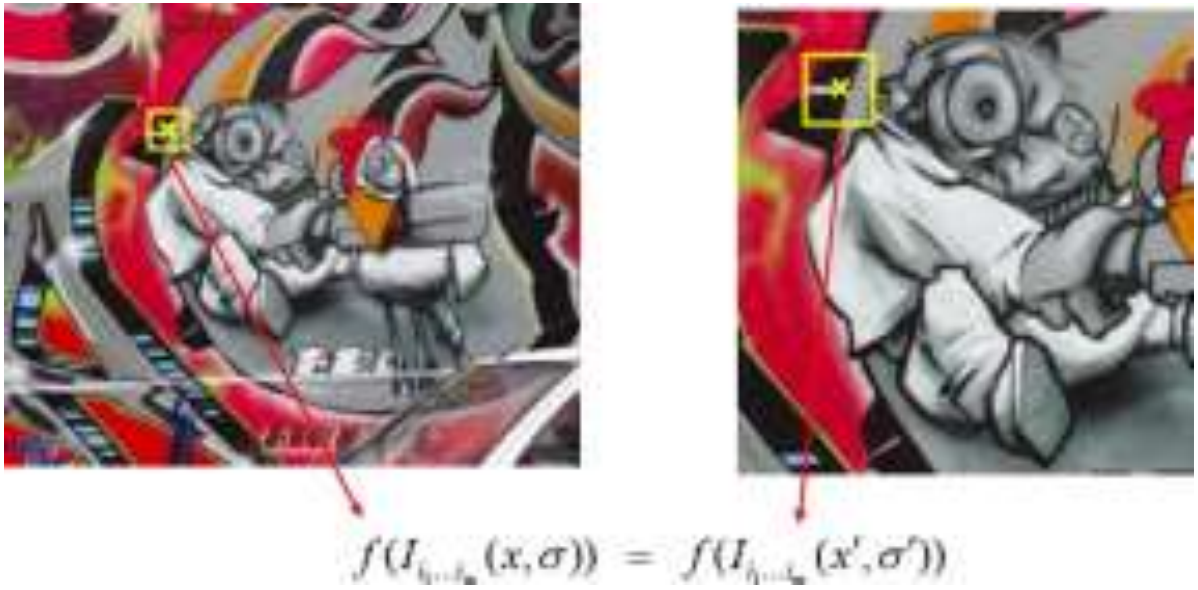
How can we **independently** select interest points in each image, such that the detections are repeatable across different scales?

- Extract features at a variety of scales, e.g., by using multiple resolutions in a pyramid, and then matching features at the same level.
- When does this work?

Scale Invariant Interest Points

How can we **independently** select interest points in each image, such that the detections are repeatable across different scales?

- More efficient to extract features that are stable in both location and scale.

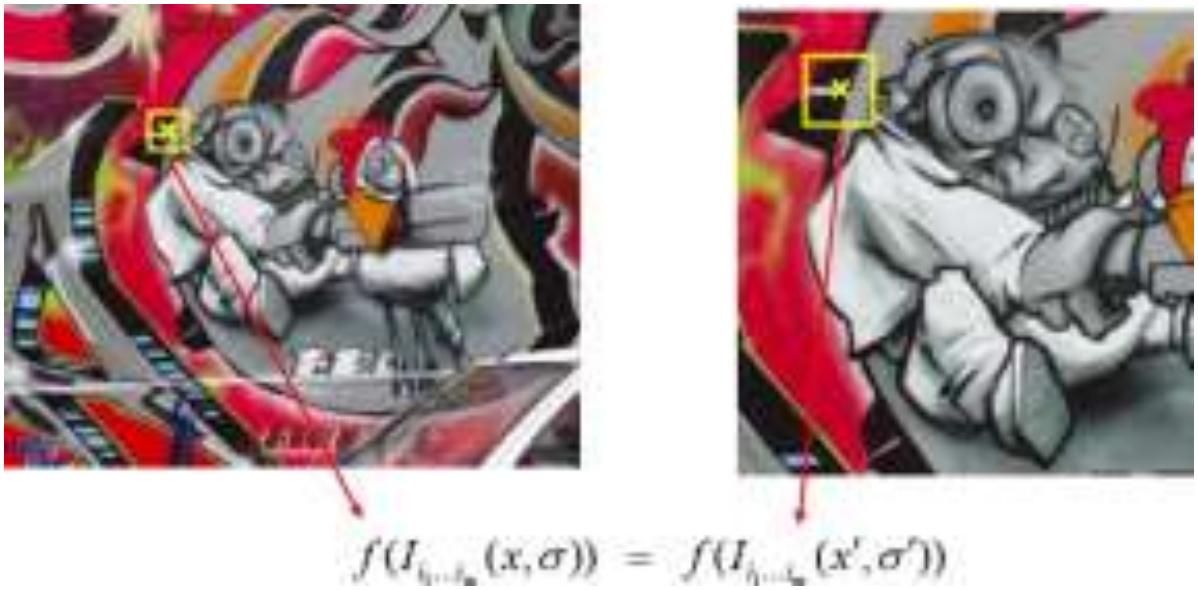


[Source: K. Grauman, slide credit: R. Urtasun]

Scale Invariant Interest Points

How can we **independently** select interest points in each image, such that the detections are repeatable across different scales?

- Find scale that gives local maxima of a function *f* in both position and scale.



[Source: K. Grauman, slide credit: R. Urtasun]

Automatic Scale Selection

Function responses for increasing scale (scale signature).

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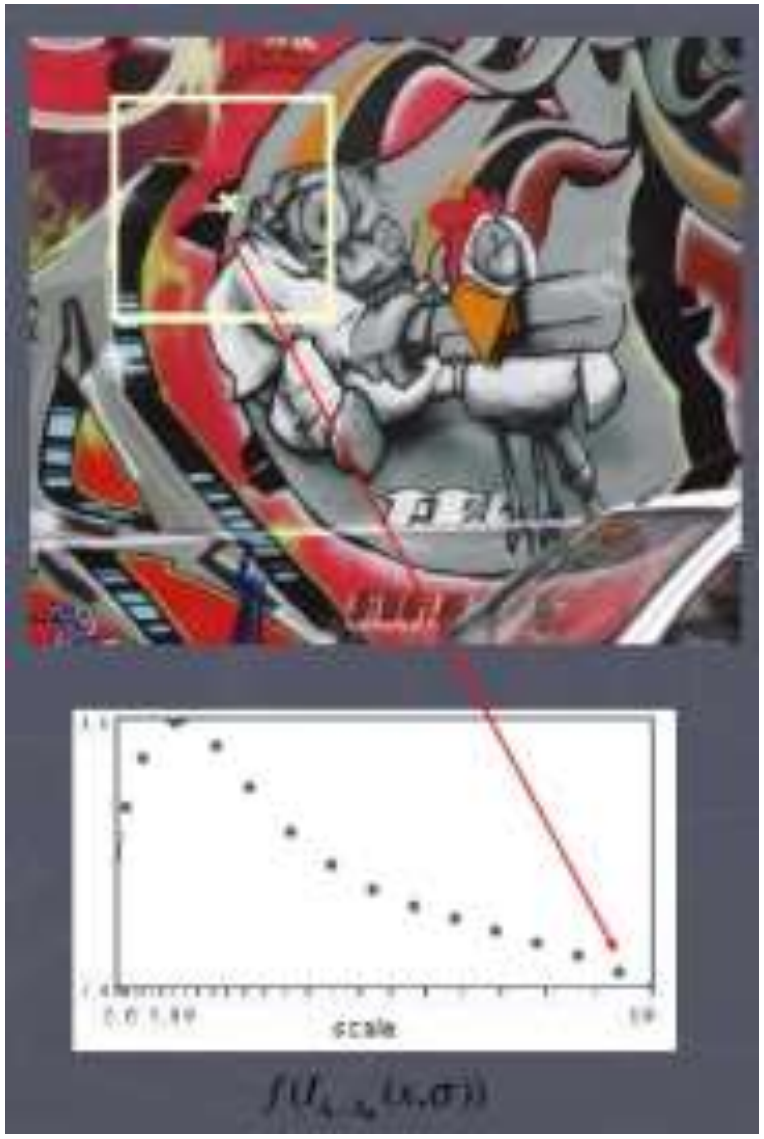

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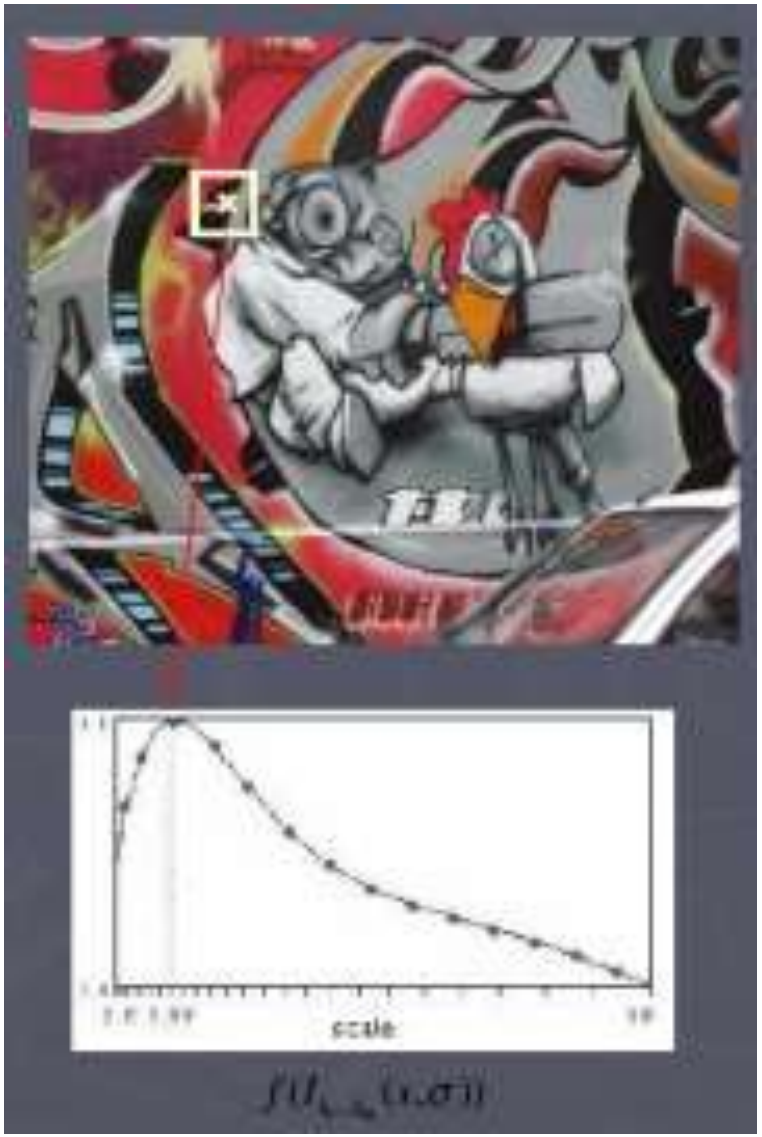
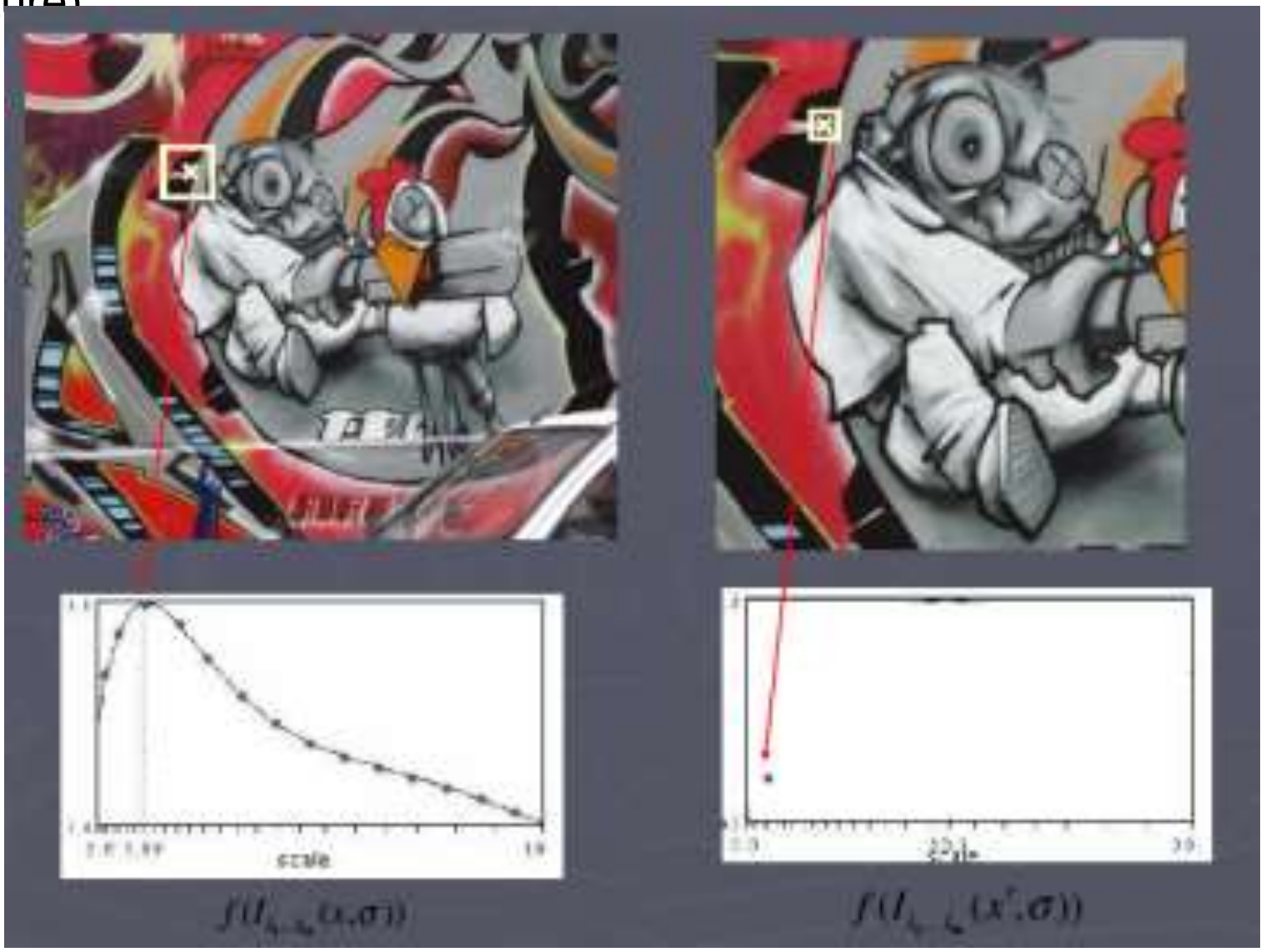


Automatic Scale Selection

Function responses for increasing scale (scale signature)



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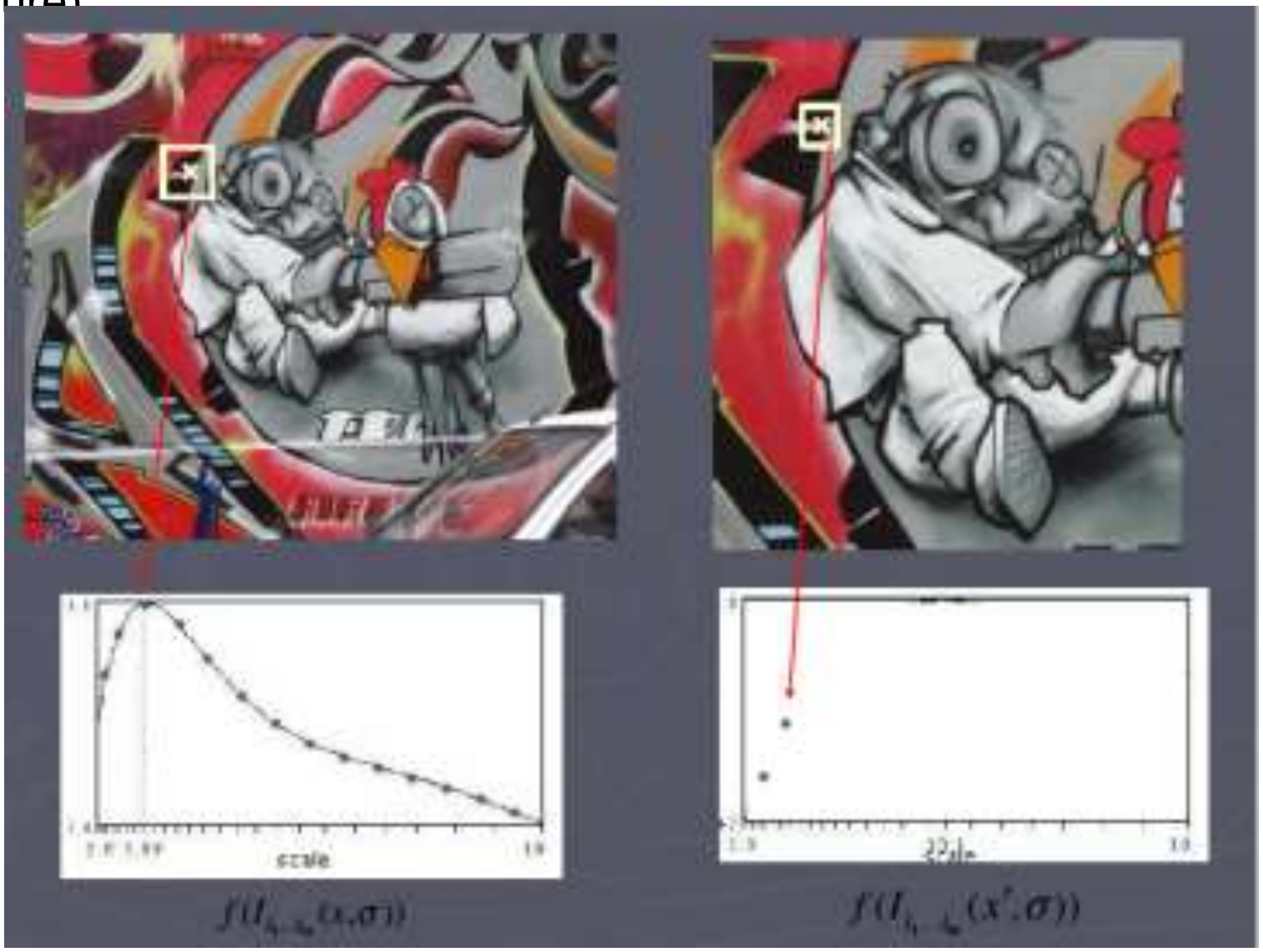

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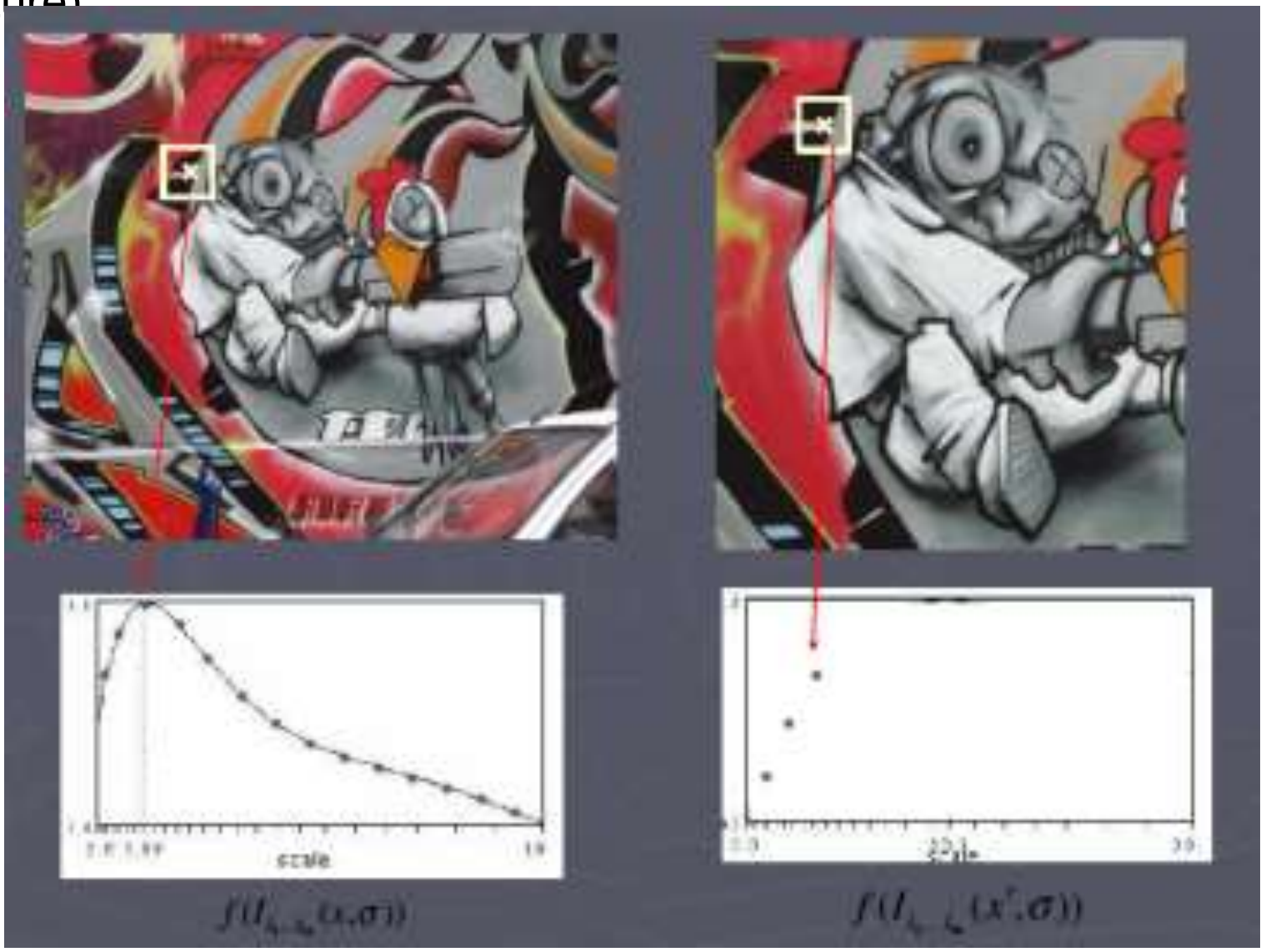

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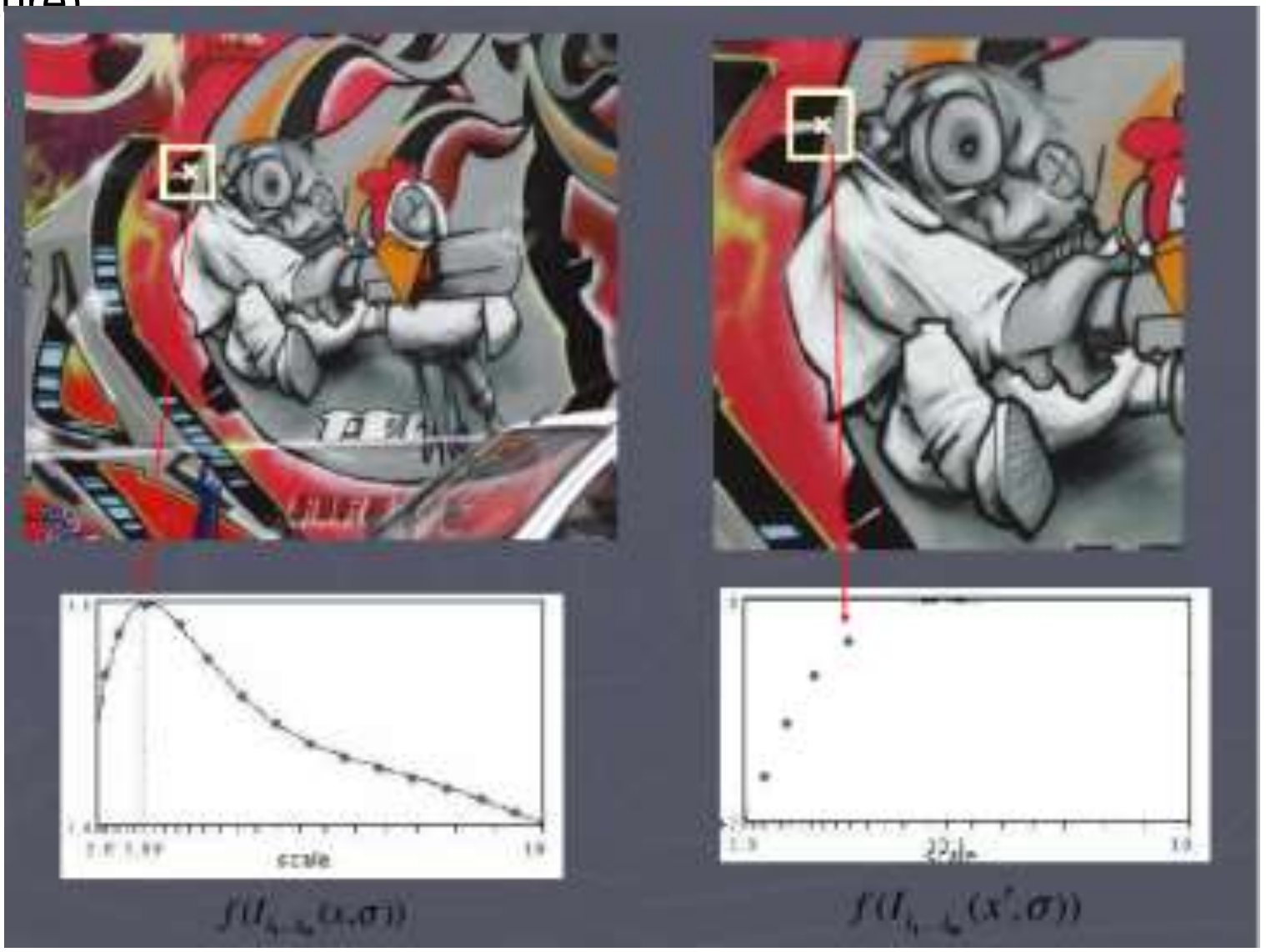

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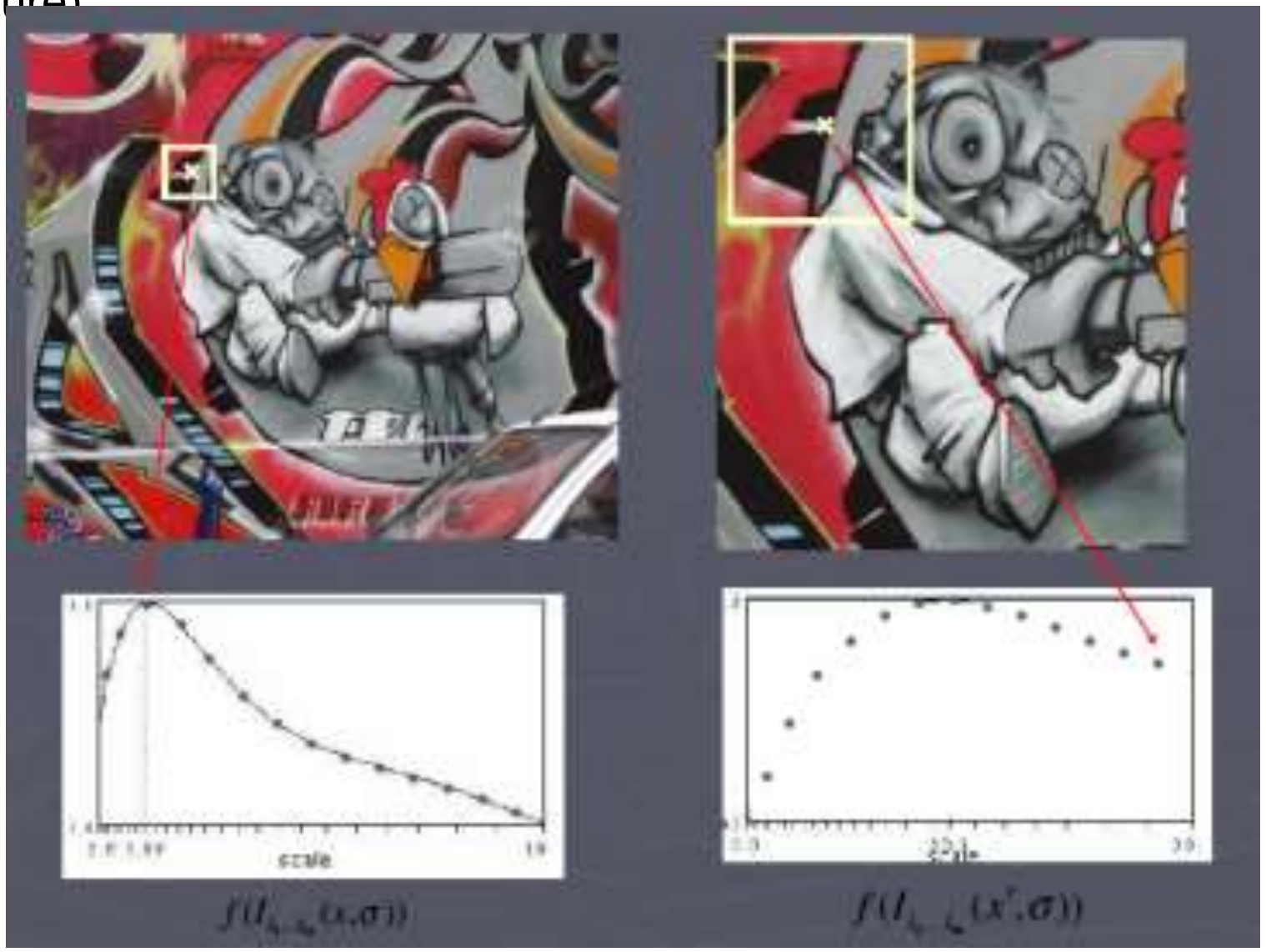

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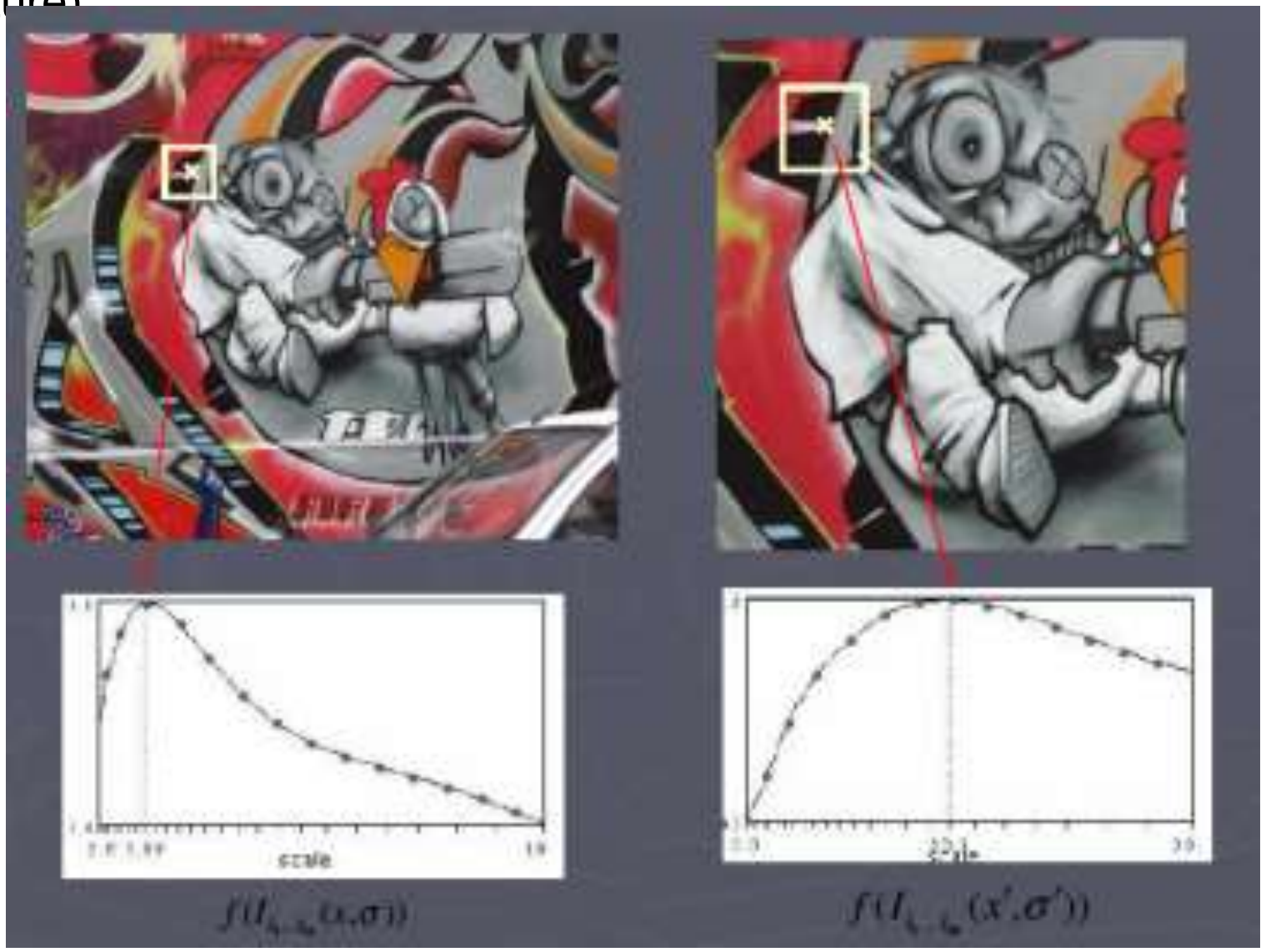


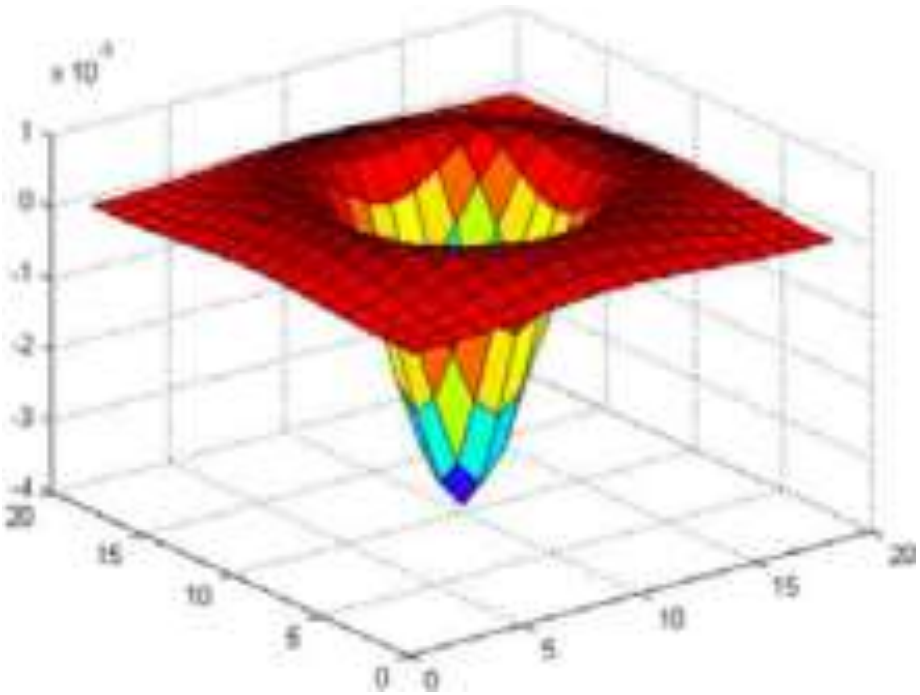
Blob Detection – Laplacian of Gaussian

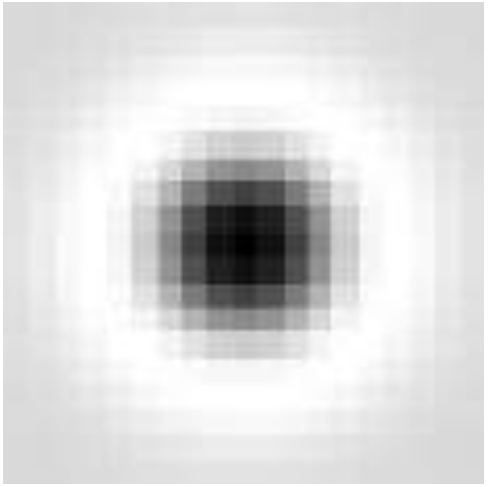
- Laplacian of Gaussian: We mentioned it for edge detection

$$\nabla^2 g(x, y, \sigma) = \frac{\partial^2 g(x, y, \sigma)}{\partial x^2} + \frac{\partial^2 g(x, y, \sigma)}{\partial y^2},$$

where g is a Gaussian

- It is a circularly symmetric operator (finds difference in all directions)
- It can be used for 2D blob detection





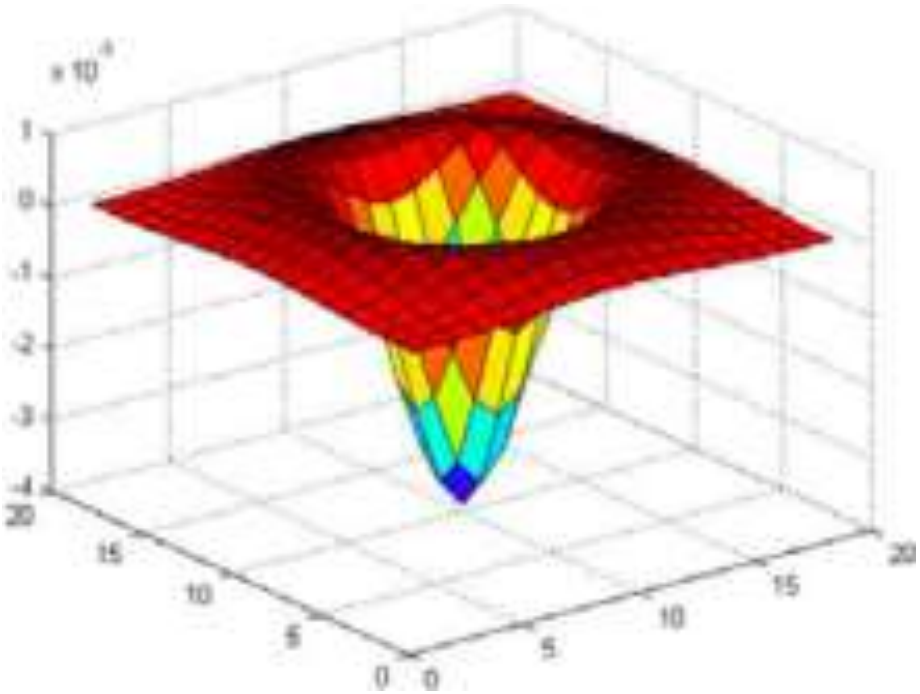
[Source: R. Gaussman, credit: D. Lowe]

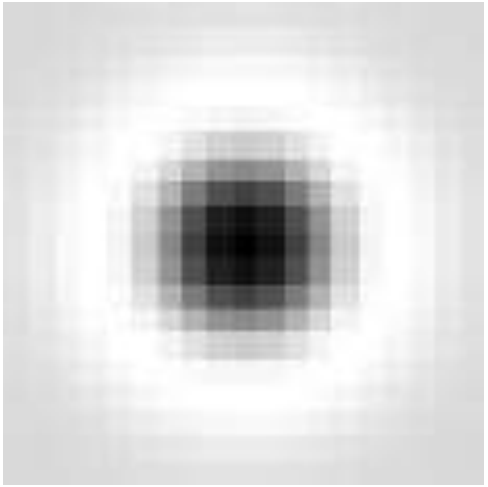
Blob Detection – Laplacian of Gaussian

- Laplacian of Gaussian: We mentioned it for edge detection

$$\mathrm{LoG}(x, y; \sigma) = \frac{1}{2\pi\sigma^4}\left(\frac{x^2+y^2-2\sigma^2}{\sigma^2}\right)\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

- It is a circularly symmetric operator (finds difference in all
- directions) It can be used for 2D blob detection

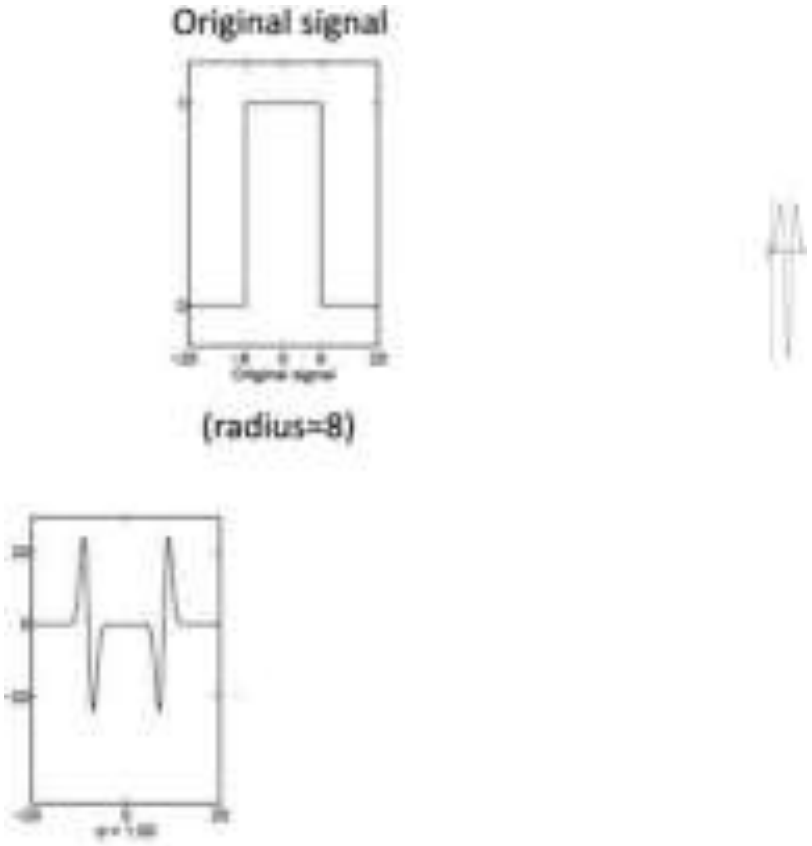




[Source: D. Forsyth, credit: D. Hoiem]

Blob Detection – Laplacian of Gaussian

- It can be used for 2D blob detection! How?



[Source: F. Flores-Mangas]

Blob Detection – Laplacian of Gaussian

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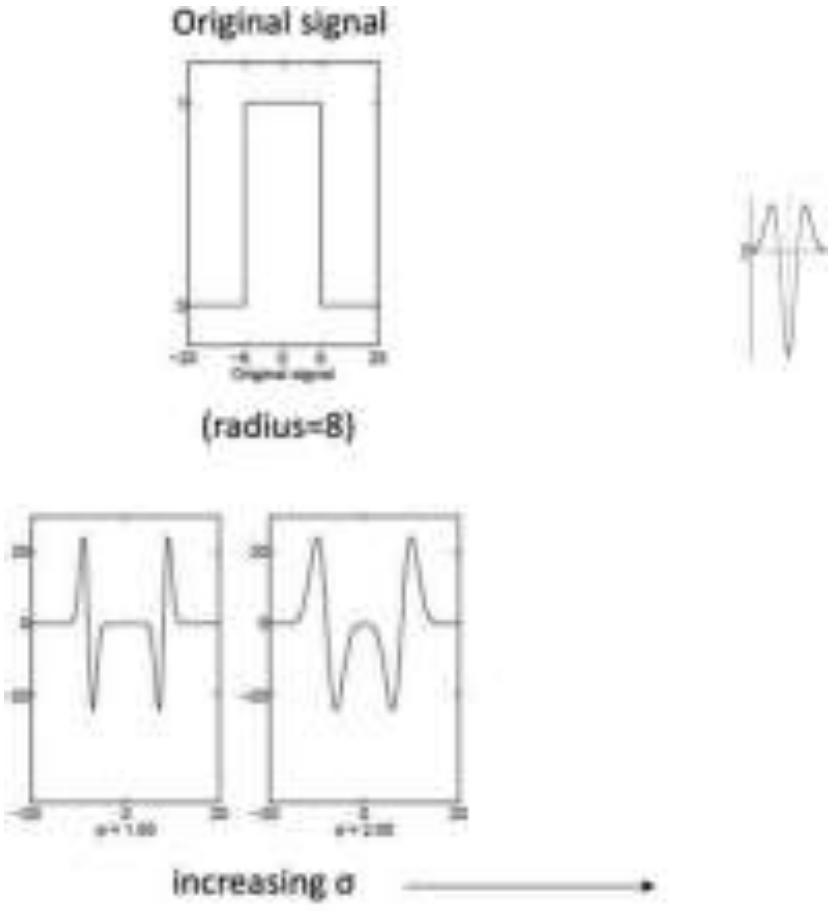
  How?



[Source: F. Flores-Mangas]

Blob Detection – Laplacian of Gaussian

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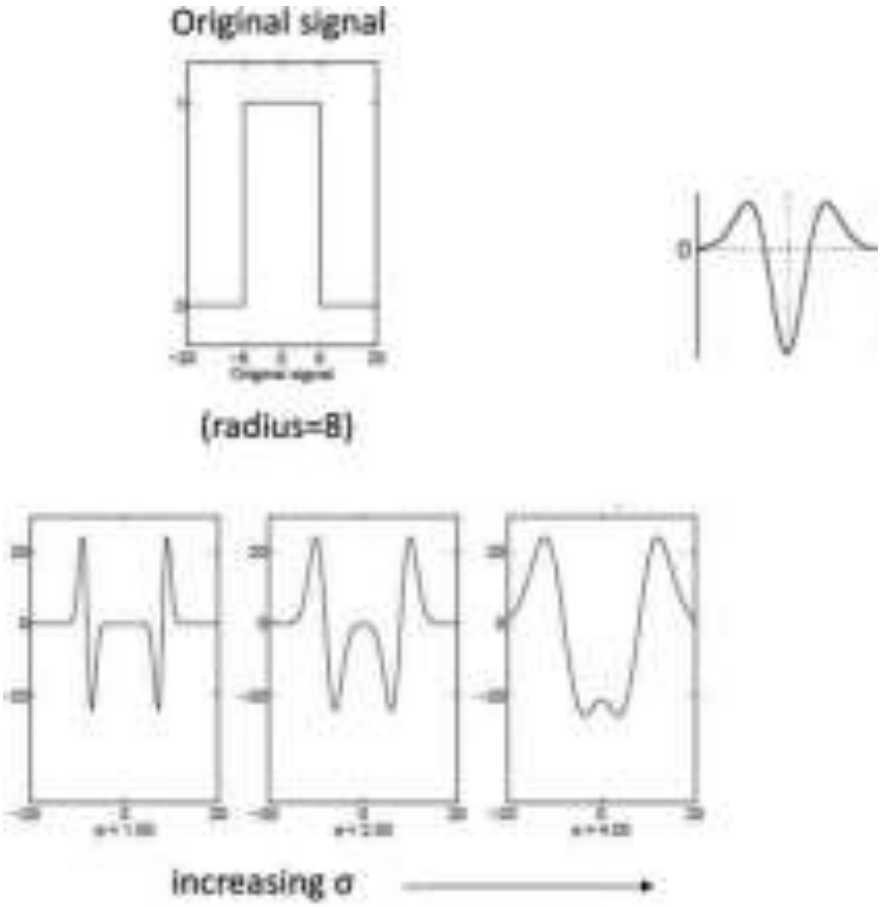

[Source: F. Flores-Mangas]

Blob Detection – Laplacian of Gaussian

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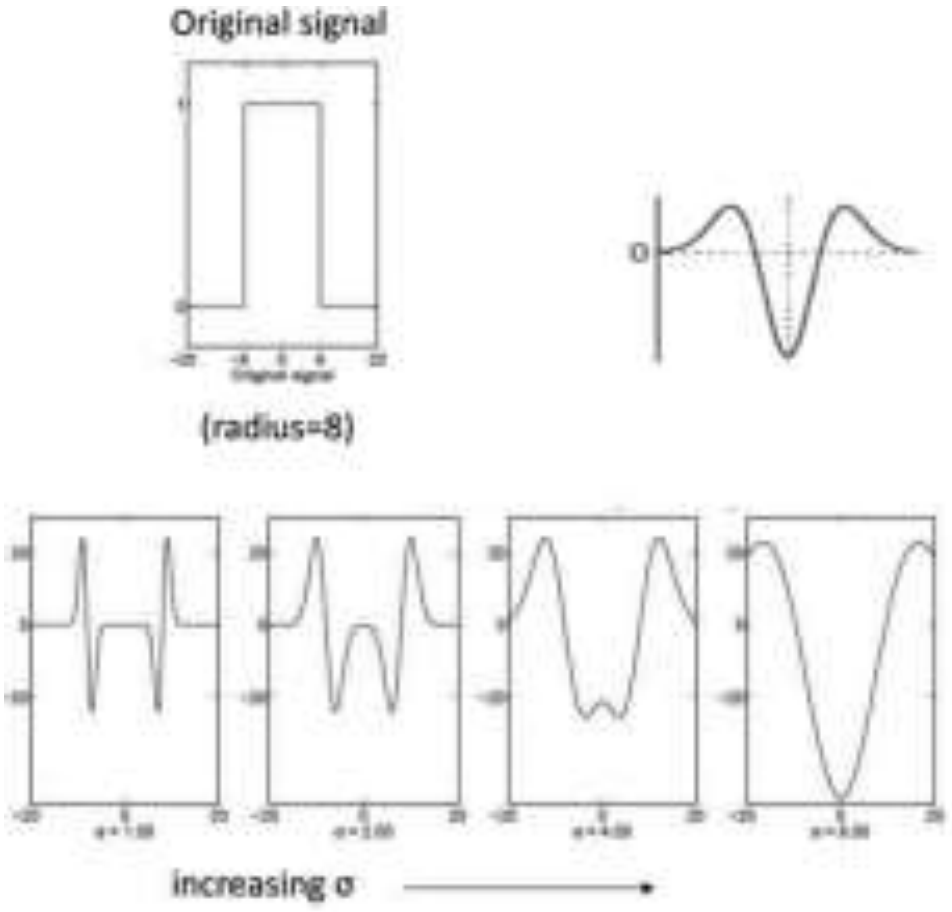

[Source: F. Flores-Mangas]

Blob Detection – Laplacian of Gaussian

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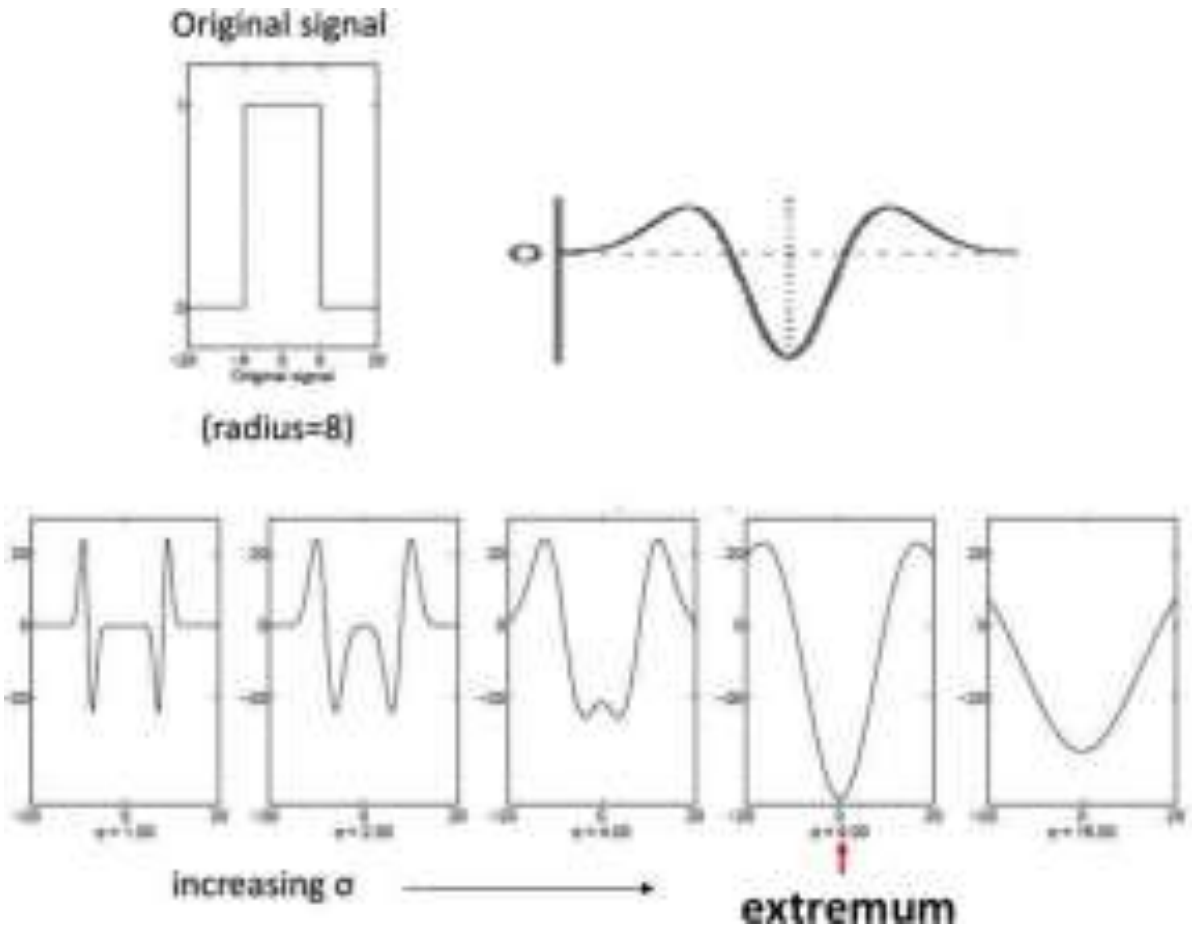
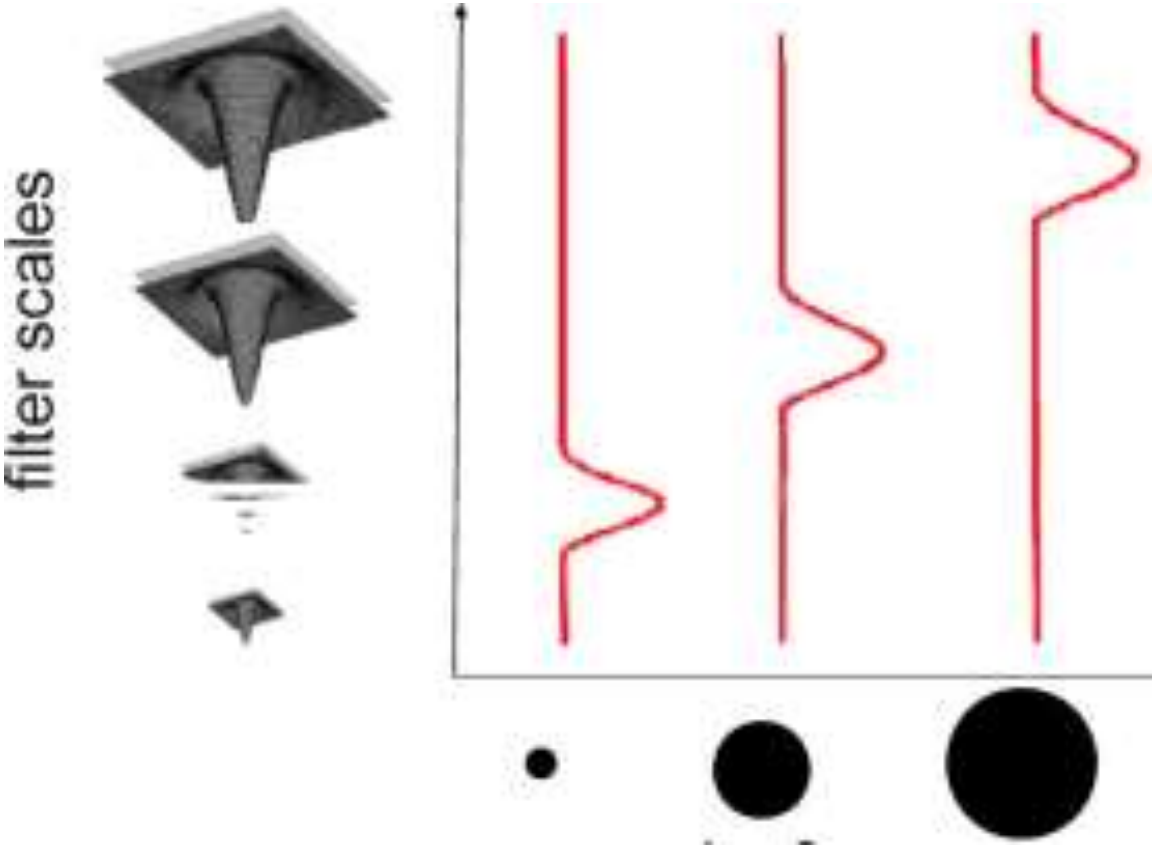


[Source: F. Flores-Mangas]

Blob Detection in 2D: Scale Selection

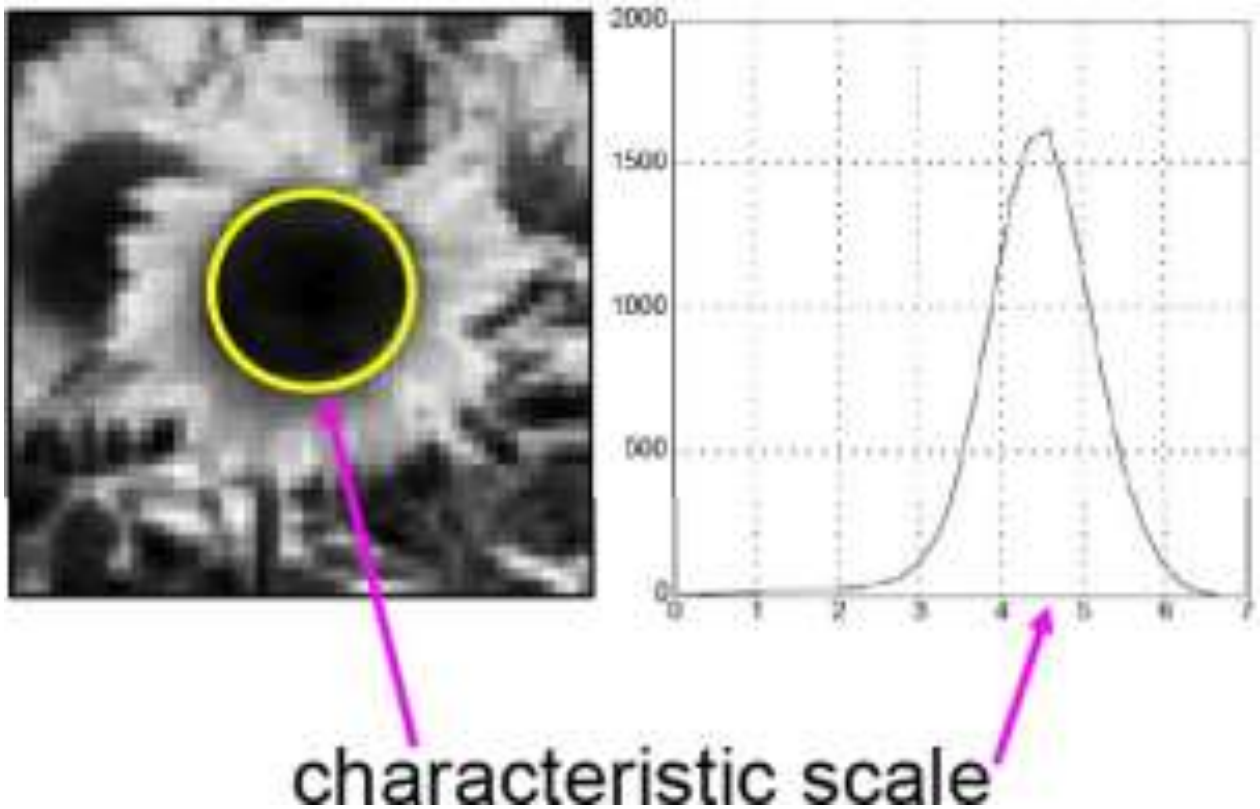
Laplacian of Gaussian = blob detector



[Source: B. Leibe, slide credit: R. Urtasun]

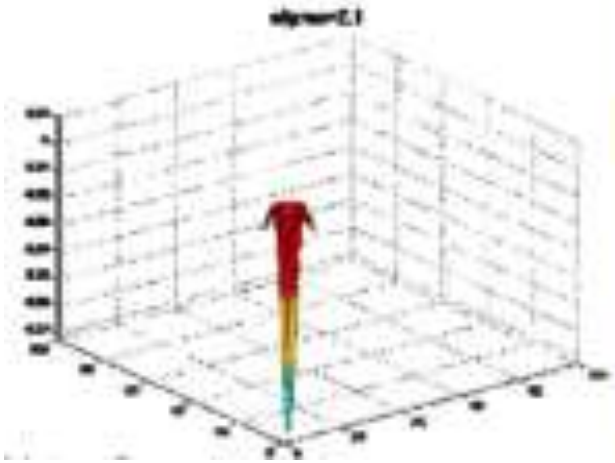
Characteristic Scale

- We define the **characteristic scale** as the scale that produces peak (minimum or maximum) of the Laplacian response



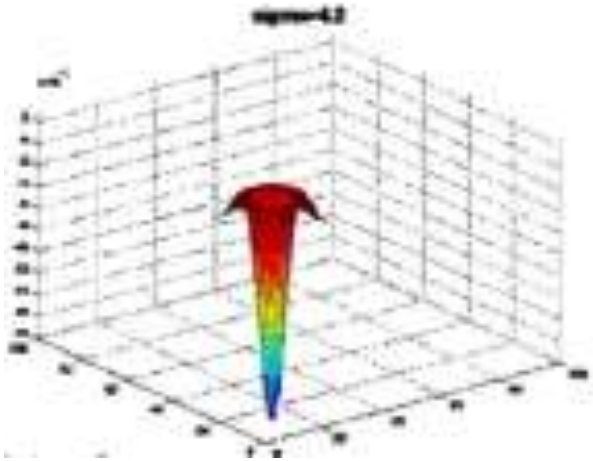
[Source: S. Lazebnik]

Example



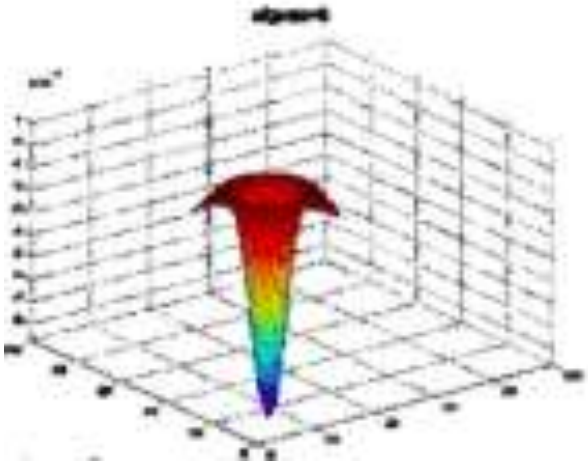
[Source: K. Grauman]

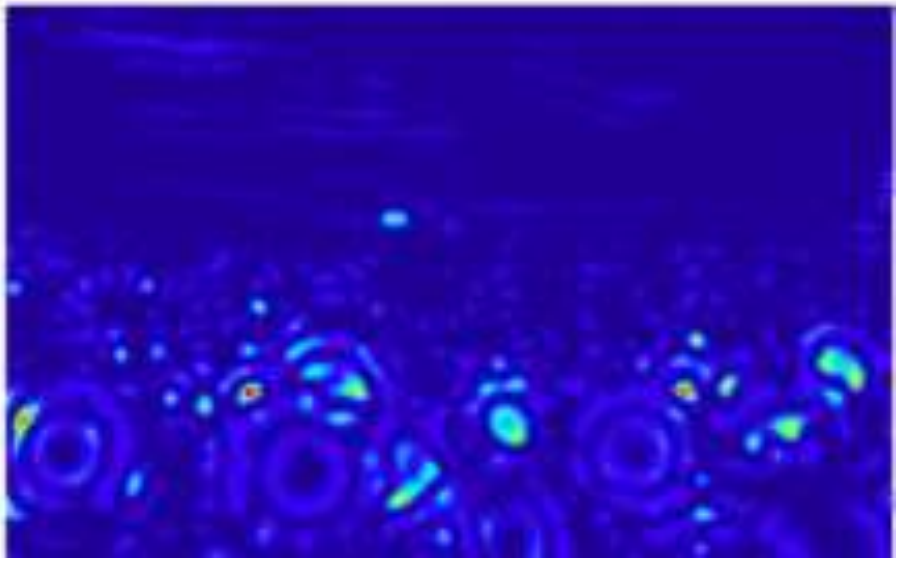
Example



[Source: K. Grauman]

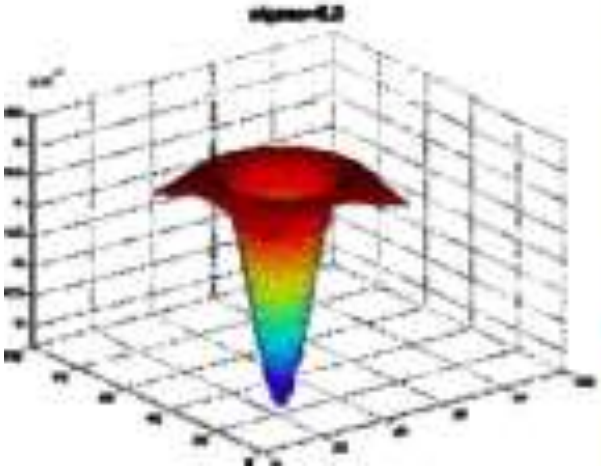
Example

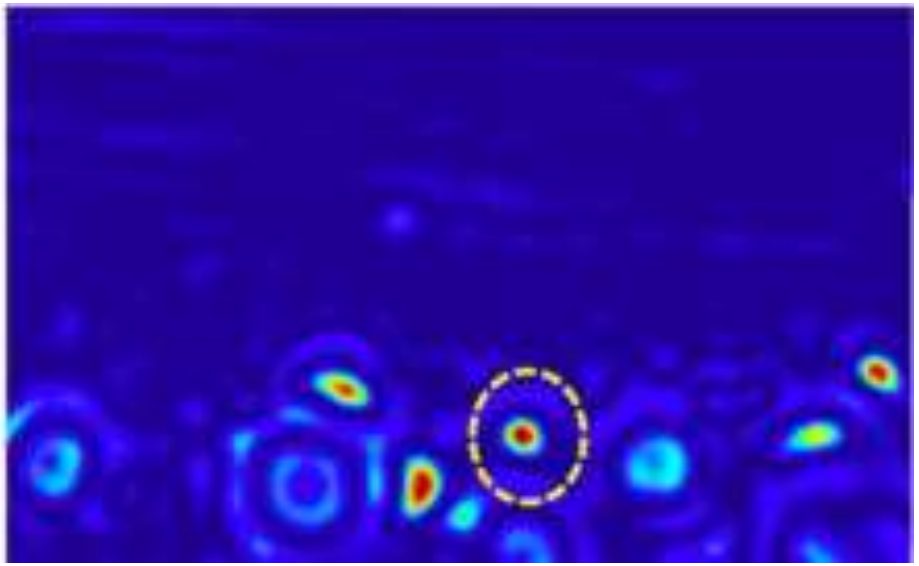




[Source: K. Grauman]

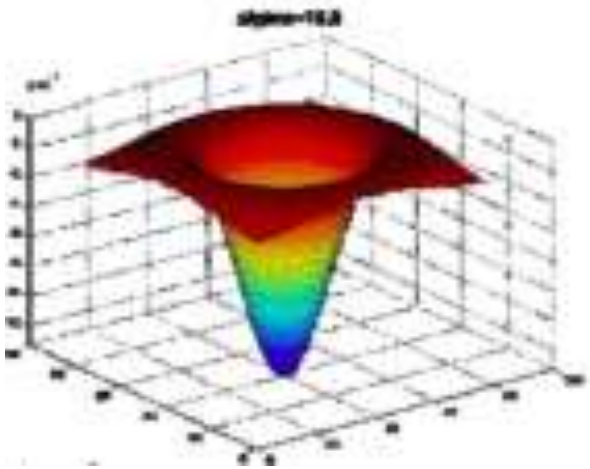
Example

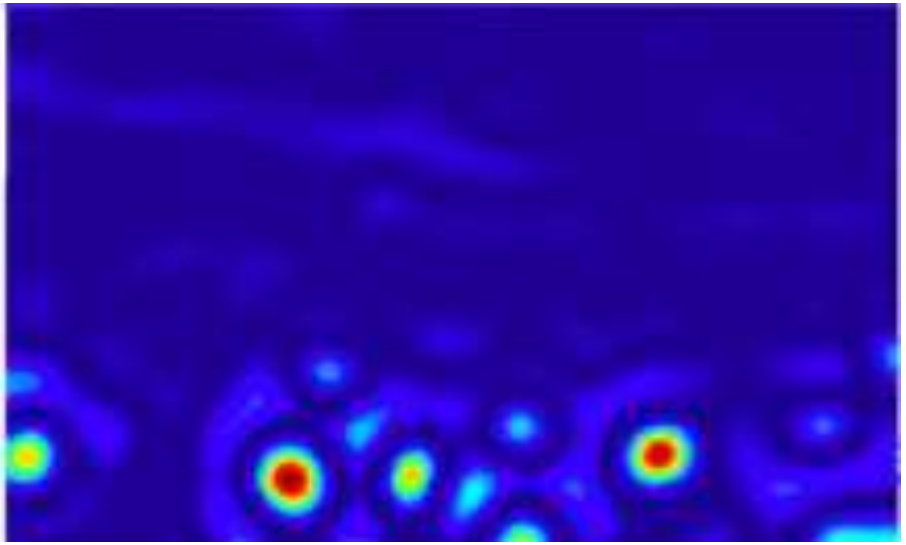




[Source: K. Grauman]

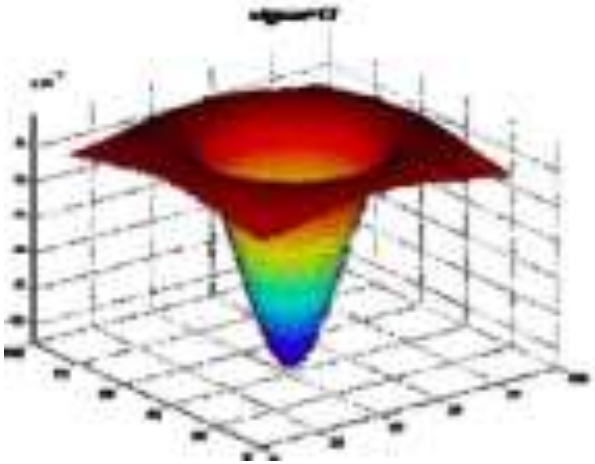
Example

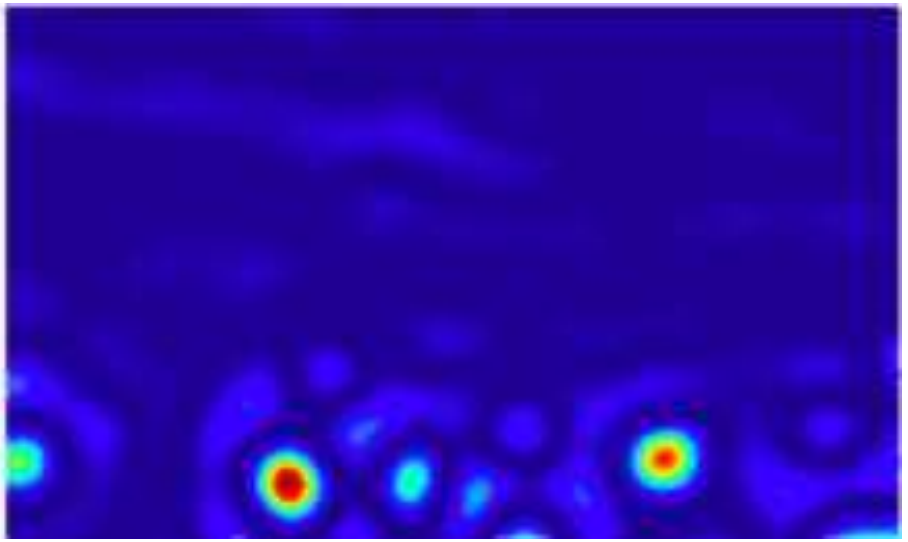




[Source: K. Grauman]

Example





[Source: K. Grauman]

Efficient implementation

Courtesy: Forsyth and Ponce

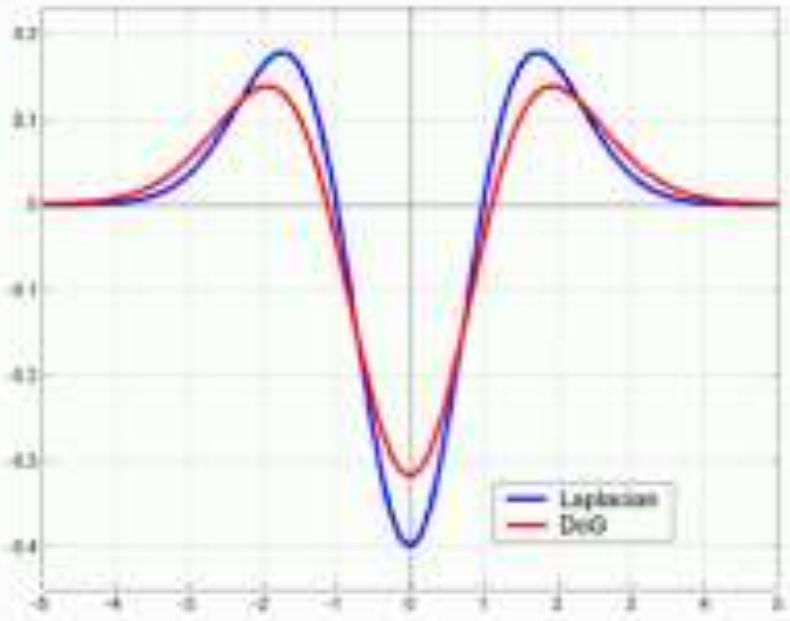
Approximating the Laplacian with a difference of Gaussians:

$$L = \sigma^2 \left(G_{xx}(x, y, \sigma) + G_{yy}(x, y, \sigma) \right)$$

(Laplacian)

$$DoG = G(x, y, k\sigma) - G(x, y, \sigma)$$

(Difference of Gaussians)



Derivation based on heat diffusion equation. For more details see Sec 3, pgs 5-6 of David Lowe, IJCV 2004 paper: https://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf

Scale Invariant Feature Transform (SIFT)

Steps in SIFT

- Step 1: Approximate keypoint location
- Step 2: Refining keypoint location
- Step 3: Assigning orientations
- Step 4: Descriptors for each keypoint

Step 1: Approximate keypoint

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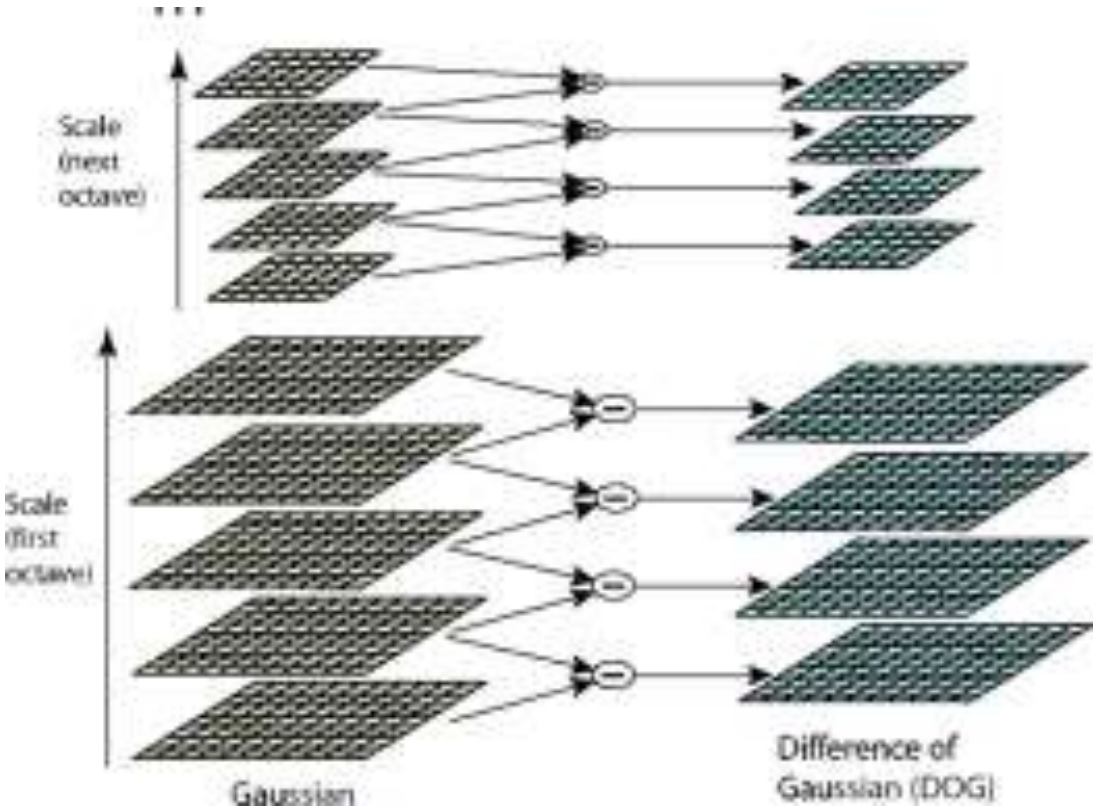


$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

Octave = doubling of σ_0. Within an octave, the adjacent scales differ by a constant factor *k*. If an octave contains *s+1* images, then $k = 2^{(1/(s))}$. The first image has scale σ_0, the second image has scale $k\sigma_0$, the third image has scale $k^2\sigma_0$, and the last image has scale $k^s\sigma_0$. Such a sequence of images convolved with Gaussians of increasing σ constitute a so-called **scale space**.

Scale = 1

Scale = 4

Scale = 16

Scale = 64

Scale = 256

http://en.wikipedia.org/wiki/Scale_space

Step 1: Approximate keypoint

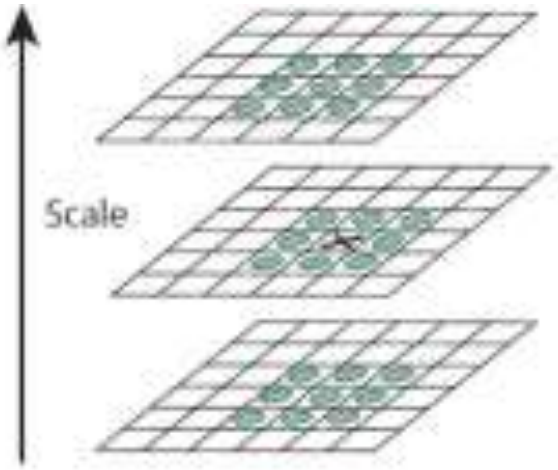


Image taken from D. Lowe, “Distinctive Image Features from Scale-Invariant Points”, IJCV 2004

Figure 2: Maxima and minima of the difference-of-Gaussian images are detected by comparing a pixel (marked with X) to its 26 neighbors in 3x3 regions at the current and adjacent scales (marked with circles).

The keypoints are maxima or minima in the “scale-space-pyramid”,
i.e. the stack of DoG images. Hereby, you get both the location as well as the scale of the keypoint.

Initial detection of keypoints

http://upload.wikimedia.org/wikipedia/commons/4/44/Sift_keypoints_filtering.jpg

Step 2: Refining keypoint location

- The keypoint location and scale is discrete – we can interpolate for greater accuracy.
- For this, we express the DoG function in a small 3D neighborhood around a keypoint (x_i, y_i, σ_i) by a second-order Taylor-series:

$$D(x, y, \sigma) = D(x_i, y_i, \sigma_i) + \boxed{\left(\frac{\partial D(x, y, \sigma)}{\partial(x, y, \sigma)}\right)^T \Delta} + \frac{1}{2}\Delta^T \boxed{\frac{\partial^2 D(x, y, \sigma)}{\partial(x, y, \sigma)^2}} \Delta;$$

$$\Delta = \begin{pmatrix} x - x_i \\ y - y_i \\ \sigma - \sigma_i \end{pmatrix}$$

Gradient vector evaluated digitally at the keypoint

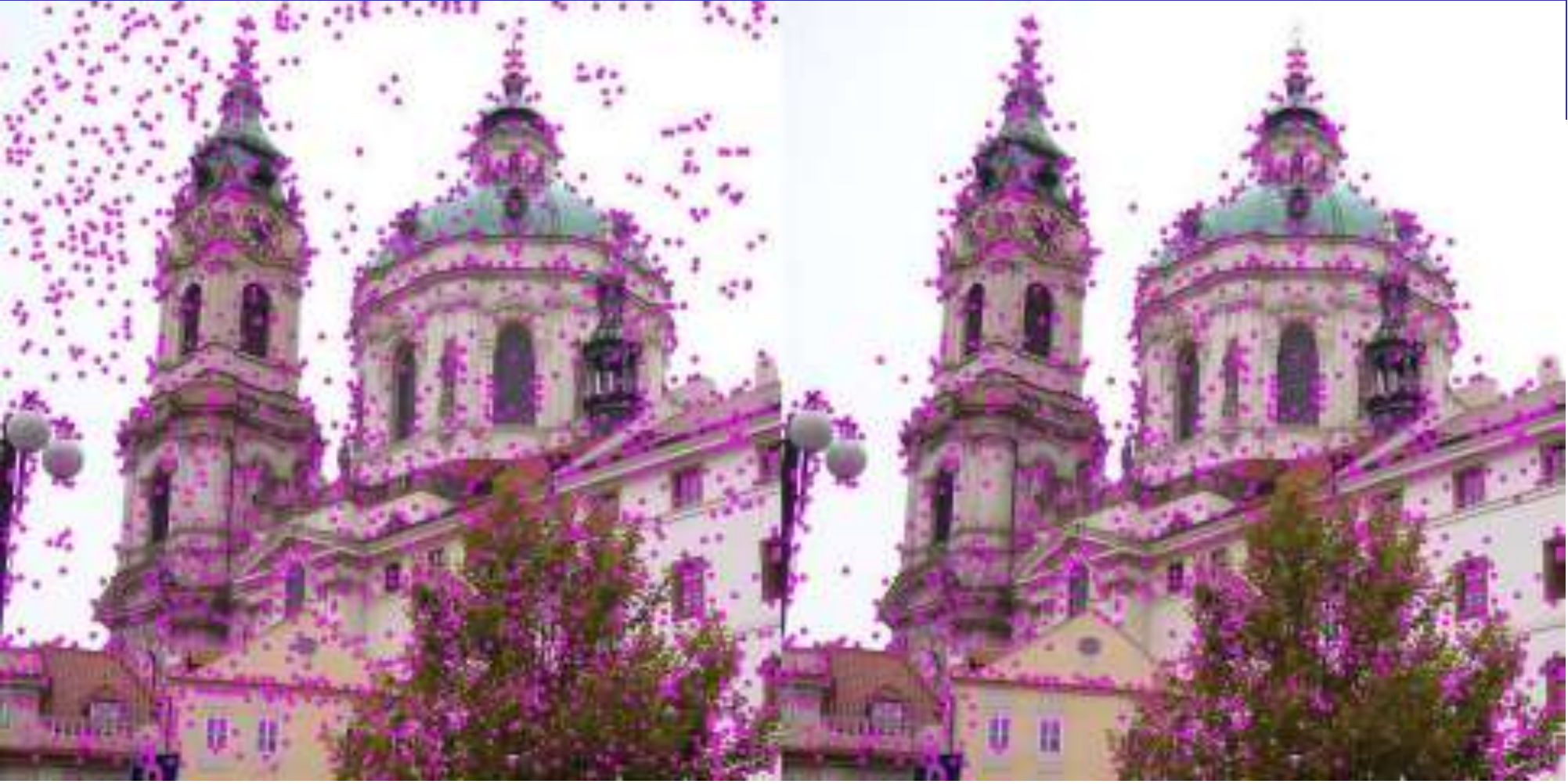
3 x 3 Hessian matrix evaluated digitally at the keypoint

Step 2: Refining keypoint location

- To find an extremum of the DoG values in this neighborhood, set the derivative of D(.) to 0. This gives us:

$$\Delta = -1 \left(\frac{\partial^2 D(x, y, \sigma)}{\partial(x, y, \sigma)^2} \right)^{-1}_{\substack{x = x_i, \\ y = y_i, \\ \sigma = \sigma_i}} \left(\frac{\partial D(x, y, \sigma)}{\partial(x, y, \sigma)} \right)_{\substack{x = x_i, \\ y = y_i, \\ \sigma = \sigma_i}}$$

- The keypoint location is updated.
- All extrema with $|D_{extremal}| < 0.03$, are discarded as "weak extrema" or "low contrast points".



Removal of low-contrast keypoints

http://upload.wikimedia.org/wikipedia/commons/4/44/Sift_keypoints_filtering.jpg

Step 2: Refining keypoint location

- Some keypoints reside on edges, as edges always give a high response to a DoG filter.
- But edges should not be considered salient points (why?).
- So we discard points that lie on edges.
- In the case of KLT tracker, we saw how to detect points lying on salient edges using the structure tensor.

Step 2: Refining keypoint location

- The SIFT paper uses the 2nd derivative matrix (called the Hessian matrix):

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$

- The eigenvalues of **H** give a lot of information about the local structure around the keypoint.
- In fact, the eigenvalues are the **maximal** and **minimal principal curvatures** of the surface D(x,y), i.e. of the DoG function, at that point.

http://en.wikipedia.org/wiki/File:Minimal_surface_curvature_planes-en.svg

Step 2: Refining keypoint location

- An edge will have **high maximal** curvature, but very **low minimal** curvature.
- A keypoint which is a corner (not an edge) will have **high maximal and minimal** curvature.
- The following can be regarded as an

$$\mathrm{Tr}(\mathbf{H}) = D_{xx} + D_{yy} = \alpha + \beta,$$

$$\mathrm{Det}(\mathbf{H}) = D_{xx}D_{yy} - (D_{xy})^2 = \alpha\beta.$$

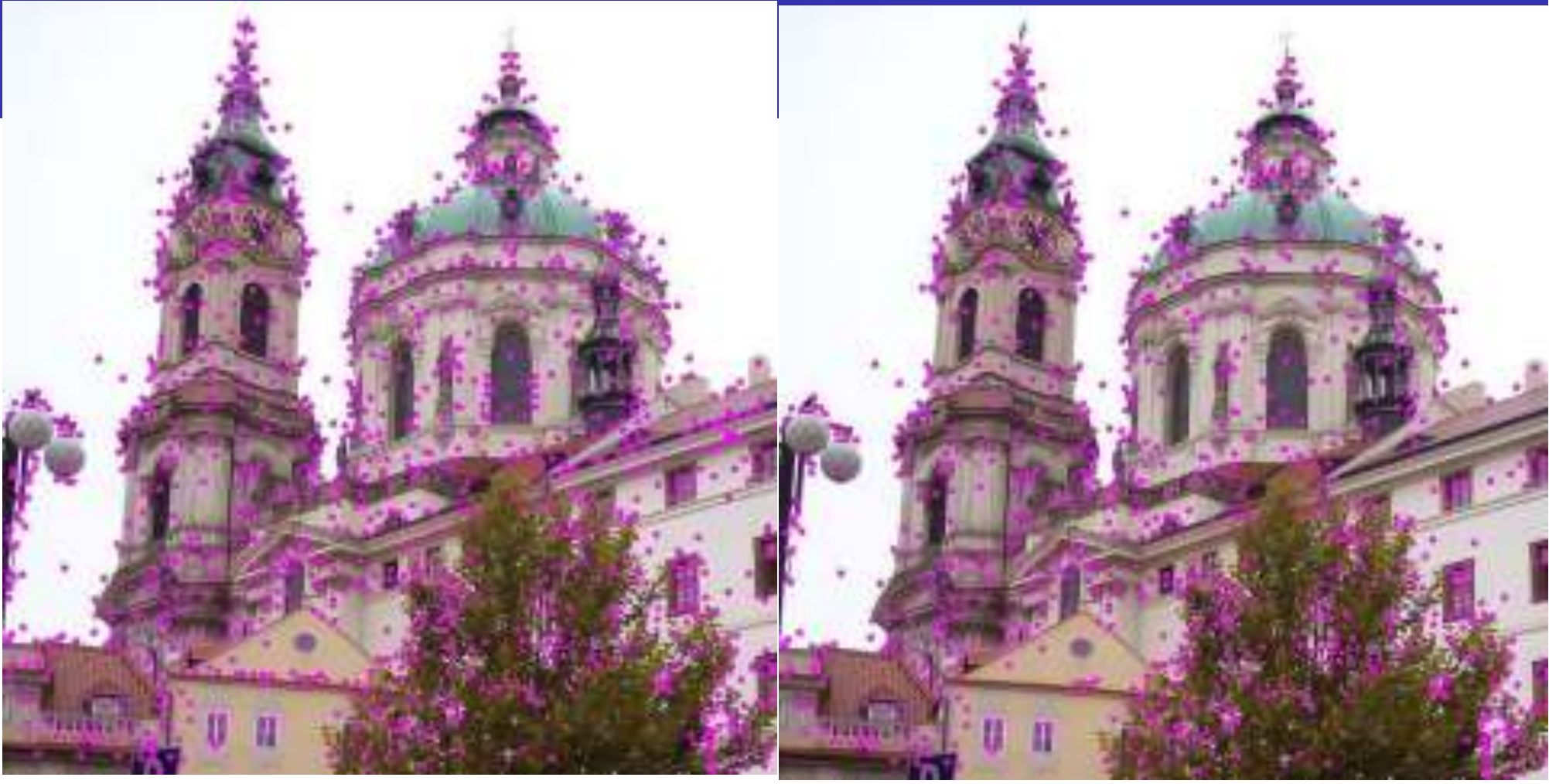
$$\boxed{\alpha = r\beta}$$

$$\boxed{\frac{\mathrm{Tr}(\mathbf{H})^2}{\mathrm{Det}(\mathbf{H})}} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r\beta+\beta)^2}{r\beta^2} = \frac{(r+1)^2}{r}$$

For a corner, should be less than a threshold (say 10).

For an edge, α >> β, leading to a large value of this measure.

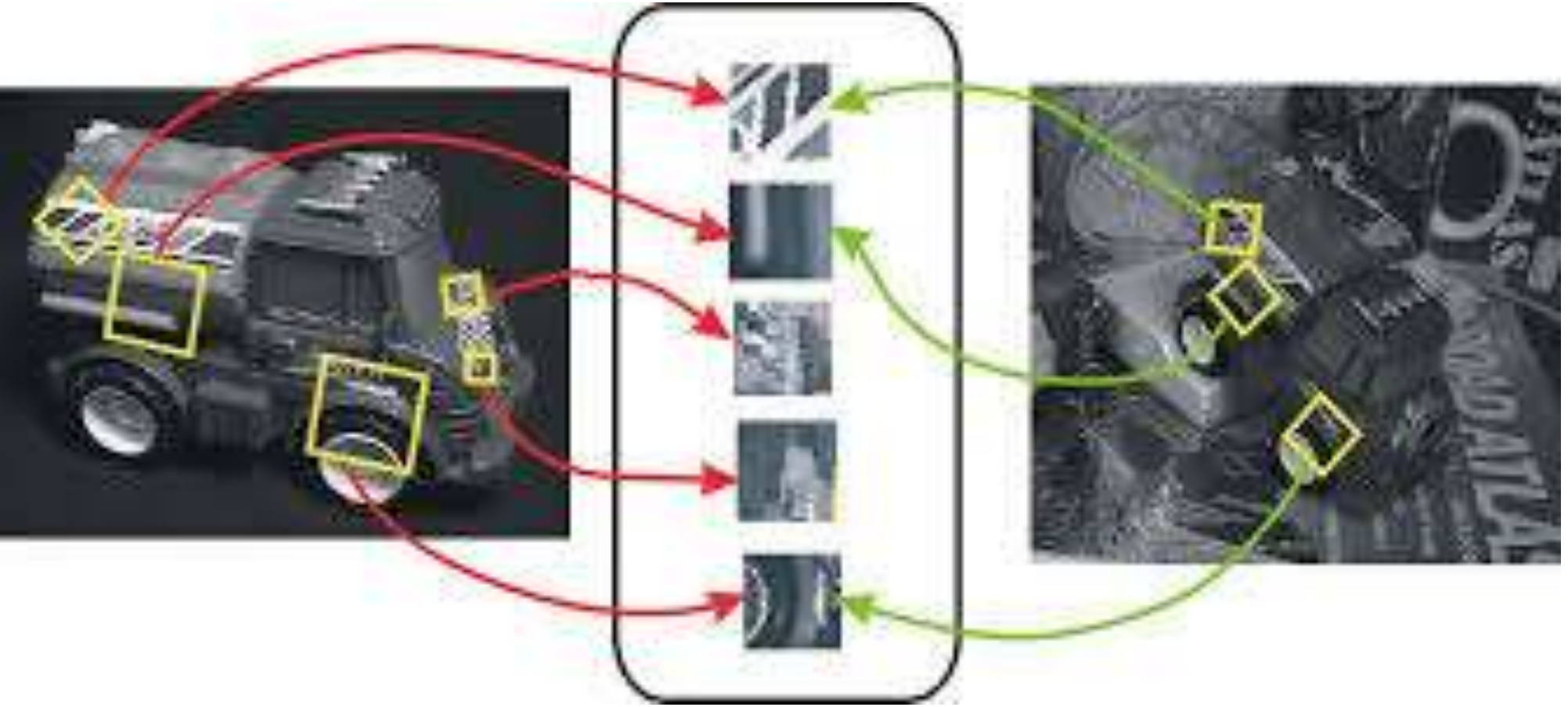
Why this measure instead of r? To save computations – we need not compute eigenvalues!



Removal of high-contrast keypoints residing on edges

http://upload.wikimedia.org/wikipedia/commons/4/44/Sift_keypoints_filtering.jpg

Step 3: Assigning orientations



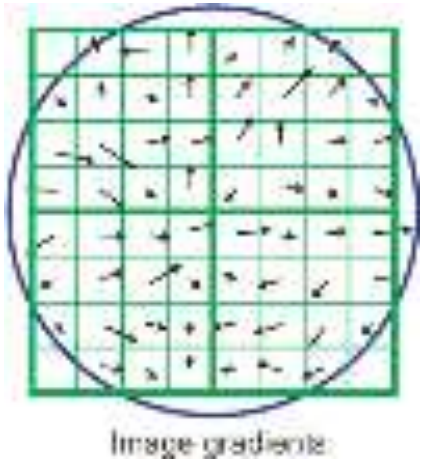
Step 3: Assigning orientations

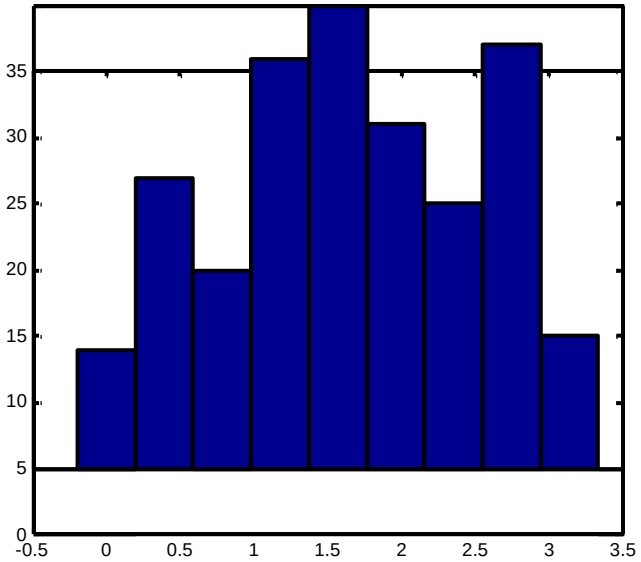
- Compute the gradient magnitudes and orientations in a small window around the keypoint – at the appropriate scale.

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2}$$

$$\theta(x, y) = \tan^{-1}((L(x, y+1) - L(x, y-1))/(L(x+1, y) - L(x-1, y)))$$

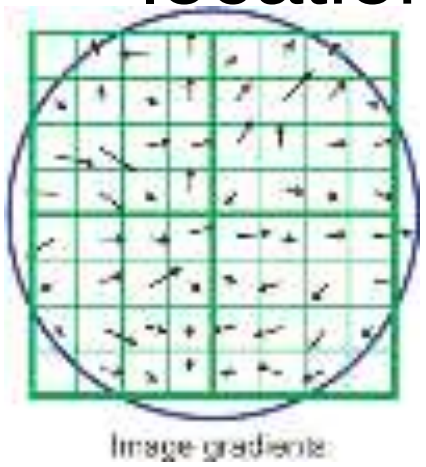


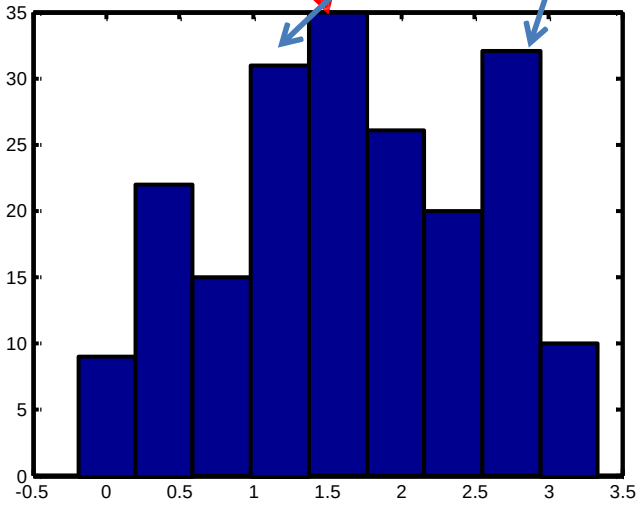


Histogram of gradient orientation – the bin-counts are weighted by gradient magnitudes and a Gaussian weighting function. Usually, 36 bins are chosen for the orientation.

Step 3: Assigning orientations

- Assign the **dominant orientation** as the orientation of the keypoint.
- In case of multiple peaks or histogram entries more than 0.8 x peak, create a **separate** descriptor for ***each*** orientation (they will all have the same scale and location).





Histogram of gradient orientation – the bin-counts are weighted by gradient magnitudes and a Gaussian weighting function. Usually, 36 bins are chosen for the orientation.

Step 4: Descriptors for each

- Consider a small region around the keypoint. Divide it into *n* x *n* cells (usually *n* = 2). Each cell is of size 4 x 4.
- Build a **gradient orientation histogram** in each cell. Each histogram entry is weighted by the *gradient magnitude* and a *Gaussian weighting function* with σ = 0.5 times window width.
- **Sort** each gradient orientation histogram bearing in mind the dominant orientation of the keypoint (assigned in step 3).

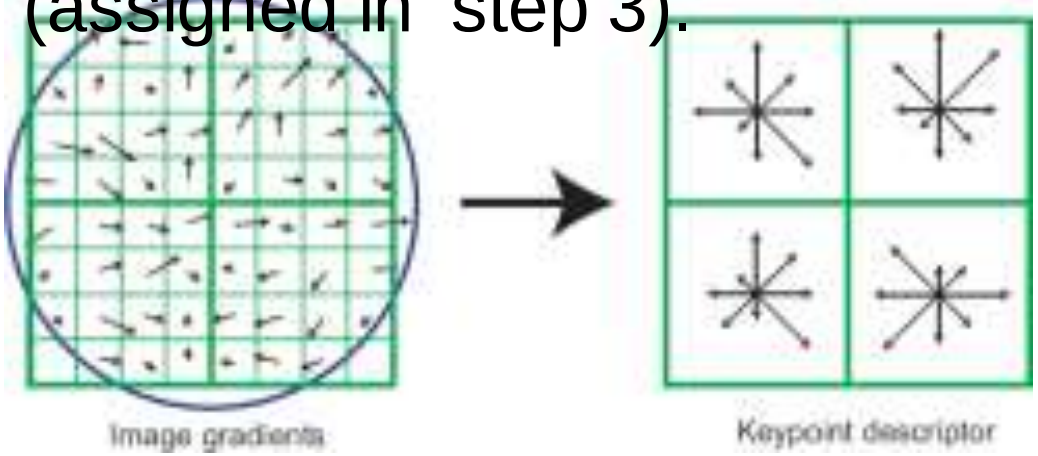


Image taken from D. Lowe, "Distinctive Image Features from Scale-Invariant Points", IJCV 2004

Step 4: Descriptors for each

- We now have a descriptor of size rn^2 if there are r bins in the orientation histogram.
- Typical case used in the SIFT paper: $r = 8$, $n = 4$, so length of each descriptor is 128.
- The descriptor is invariant to rotations due to the sorting.

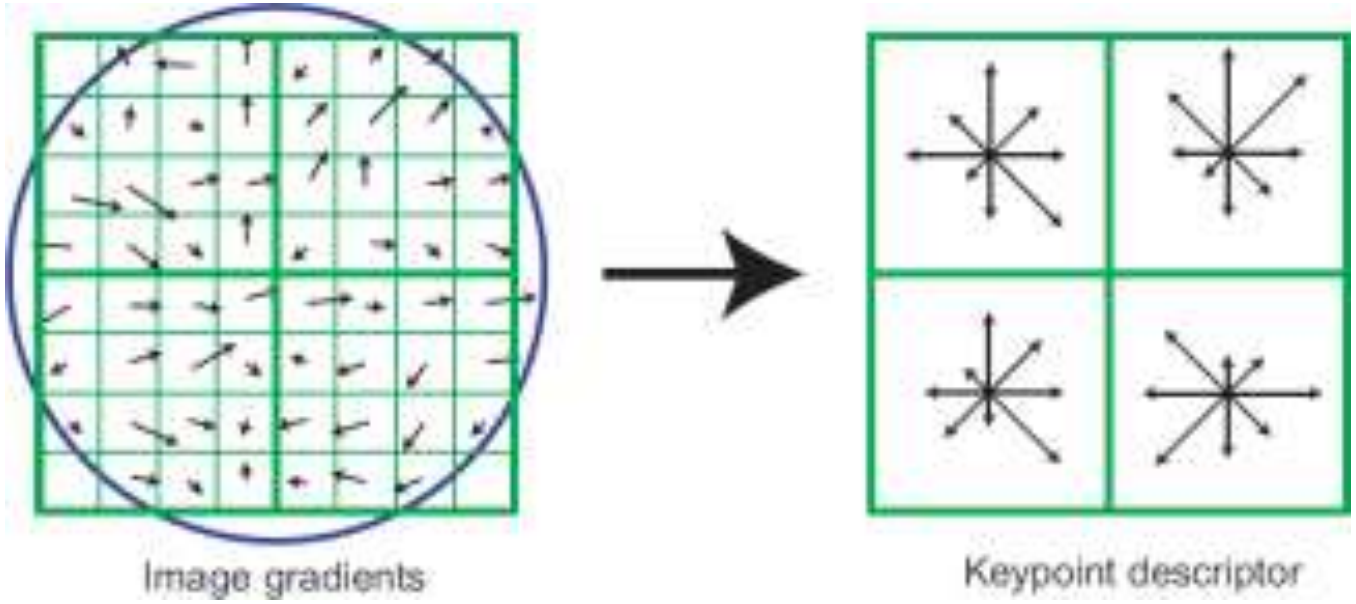


Image taken from D. Lowe, “Distinctive Image Features from Scale-Invariant Points”, IJCV 2004

Invariances of SIFT descriptor

- Scale, translation and rotationally invariant

- Regarding illumination
  - The SIFT descriptor (so far) is not illumination invariant – the histogram entries are weighted by gradient magnitude.
  - Hence the descriptor vector is normalized to unit magnitude. This will normalize scalar multiplicative intensity changes.
  - Scalar additive changes don't matter – gradients are invariant to constant offsets anyway.
  - Not insensitive to non-linear illumination